DIVISION OF AGRICULTURE
RESEARCH & EXTENSION
University of Arkansas System

RESEARCH INCENTIVE GRANTS

1. Project Title: Investigating transpiration responses of soybean genotypes to vapour pressure deficit for enhanced water use efficiency and drought tolerance

2. Principal Investigator: Elvis Elli, Assistant Professor, Crop Physiology, CSES
Co-PIs: Samuel Fernandes, Assistant Professor, Agricultural Statistics, CSES
Caio Canella Vieira, Assistant Professor, Soybean Breeding, CSES

3. Project performance dates: 04/2024-02/2026

4. Project Summary/Abstract
There is an urgent need to improve water use efficiency (WUE) of both rainfed and irrigated cropping systems, and breeding crop varieties with higher WUE is indicated as part of the solution. However, little progress has been made in developing genetic materials expressing improved physiological traits for drought, including WUE. This proposal aims to enhance our fundamental understanding of key physiological mechanisms underlying WUE by quantifying transpiration responses of diverse soybean accessions regarding WUE plasticity to increased vapour pressure deficit (VPD) levels. Our overarching hypothesis is that genotypes with expected WUE plasticity to the environment would limit their transpiration rates under high levels of VPD. In contrast, accessions with expected WUE stability would continue to increase the transpiration rate as VPD increases. Transpiration responses to VPD will be quantified using controlled-environment chambers and a set of 16 portable balances linked to an Arduino microcontroller board to record real-time values of water losses by transpiration. Soybean genotypes will include accessions with expected WUE plasticity and accessions with expected WUE stability based on the findings of Chamarthi et al. (2022). Transpiration rates (mg H_2O m^{-2} s^{-1}) for each genotype will be regressed against VPD levels using linear and linear plateau models to investigate whether genotypes would limit their transpiration rates under high VPD levels. With this proposal, we expect to advance fundamental knowledge of soybean responses to VPD, which would facilitate the conceptualization of genetic solutions to enhance yields under current and future climate change scenarios.

5. Project Narrative

***Introduction*:** Approximately 80 countries with 40% of the world's population already suffer from water scarcity (Hamdy et al., 2003; Vadez et al., 2023). Genetic solutions to minimize crop yield losses due to water deficit will be critically important in the context of climate change. While irrigation has played a critical role in increasing soybean seed yields and decreasing year-to-year yield variability in many regions of the US, a critical challenge in the future will be meeting water requirements as resources are becoming more and more scarce (Reba et al., 2017). Enhanced water use efficiency (WUE) in soybeans is among the most relevant adaptative traits for increased drought tolerance (Condon et al., 2004). However, high WUE may be detrimental to crop growth in well-watered, highly productive environments. Ideally, soybean genotypes

would be plastic for WUE, i.e., capable of altering WUE depending on the environment (Chamarthi et al., 2022). Recent research has identified maturity group IV soybean accessions with associated plasticity and stability of carbon 13 ratio, a surrogate measure for water use efficiency (Chamarthi et al., 2022). WUE patterns are generally associated with how plants alter their transpiration rates in response to vapour pressure deficit (VPD). VPD is the driving force of plant transpiration and is expected to increase under climate change due to rising temperatures (Pan et al., 2015). This proposal seeks to explore the hypothesis that genotypes with expected WUE plasticity would limit the transpiration rate under high levels of VPD, while accessions with expected non-adaptative WUE plasticity would continue to increase the transpiration rate as VPD increases. Our overarching objective is to quantify the transpiration response of diverse soybean accessions in terms of WUE plasticity to increased VPD levels.

Rationale and Significance: WUE is considered an instrumental plant trait in improving soybean yields under drought conditions, but the exact nature of this trait is still unclear and this knowledge gap needs to be addressed. In fact, raising WUE is an urgent need for both irrigated and rainfed cropping systems (Condon et al., 2004). Little progress has been made in developing genetic materials expressing specific physiological traits for drought (Sinclair et al., 2017). One possible reason is the lack of understanding of key physiological mechanisms underlying drought-tolerant cultivars. Higher WUE could be achieved by increasing plant photosynthetic efficiency, decreasing stomatal conductance (and consequently plant transpiration), or a combination of both (Condon et al., 2004). VPD and plant transpiration follow a diurnal pattern, with the lowest values at sunrise and increasing to a maximum value at around 15:00. However, the increase in transpiration has limits, and a limiting maximum transpiration rate is commonly reached at a VPD of around 2.0-3.0 kPa depending on the cultivar or species. In recent years, there has been progress in understanding a limited-transpiration trait under elevated VPD that could potentially increase plant growth and yield under water-deficit conditions. This trait has been particularly desirable due to saving water use early in the growing season so that water is conserved for use later in the season to sustain plant growth rate during seed set and fill stages (Raymundo et al., 2023). The limited-transpiration trait has been successfully identified in several major crops such as soybean, maize, sorghum, and peanut. For soybean, Fletcher et al. (2007) found that the ‘slow-wilting’ genotype PI 416937 expressed the limited transpiration trait under high VPD conditions. Sinclair et al. (2010) estimated through model simulations that all soybean regions in the US regions would have a probability of yield gain of >75% as the benefit of the limited transpiration trait. This study would enhance our understanding of key physiological mechanisms underlying WUE, facilitating the conceptualization of genetic solutions to improve soybean yields under current and future climate change scenarios.

Approach: Transpiration response to increasing VPD will be measured using controlled-environment chambers. The experiment will have a random block design with a factorial of eight soybean genotypes, five VPD levels (~1, 1.5, 1.9, 2.3, and 3 kPa), and four replications. The genotypes will include accessions with expected WUE plasticity and accessions with expected WUE stability (Chamarthi et al., 2022). Soybean genotypes will be planted in 10-cm diameter, 40-cm long pots constructed of polyvinyl chloride pipe, sealed at the bottom and top with end caps. A 2-cm diameter hole is present in the top through which the plant shoot will emerge and through which water will be applied. Potential water losses by soil water evaporation are expected to be negligible. Individual pot tare weights will be recorded to monitor the pot mixture

water content. Plants will be well-watered until the measurement date (beginning flowering, R1) by watering to 70% of pot-capacity weight as needed to keep soil water contents from falling below 50% of the pot-capacity weight (King et al., 2009). The growth chamber will be maintained at a constant temperature of 28°C with a light cycle from 06:00 am to 22:00 h, photosynthetically active radiation at the top of the plants of approximately 600 μmol m^{-2} s^{-1}, and relative humidity of approximately 60% until the measurement date. Portable balances will be installed at the beginning of the experiment under each pot with soybean plants. Those balances will be linked to an Arduino microcontroller and datalogger system which will measure and record individual pot weights every minute during the measurement day. When the VPD treatment is initiated, the pot weight between subsequent weighing times is assumed to be water lost through transpiration. Measurements will start at the lowest VPD treatment and move to the increasing VPD treatments (Fletcher et al., 2007). Additional experiment runs will be performed with a randomized order of VPD treatments to ensure this factor does not affect the overall transpiration response. Whole-plant transpiration rates will also be analyzed per unit leaf area (mg H_2O m^{-2} s^{-1}). After transpiration measurements, the plant leaf area will be determined destructively by passing the leaves through an area meter. Transpiration rates for each genotype will be regressed against VPD levels (Fletcher et al., 2007). Linear and linear-plateau regression models will be evaluated using the lme4 R package (Bates et al., 2014; R Development Core Team, 2021). Best-fit models will be selected based on the highest R^2.

Work Plan and Timeline	Year 1				Year 2				Year 3
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
Programming and setting up the portable balance and Arduino datalogger system									
Growth chamber experiments & data collection									
Data analysis & statistics									

6. How will this project be leveraged to apply for extramural funding?

This proposal will generate preliminary results for a NIFA proposal with the objective of designing multiple genetic strategies to maximize soybean yields in the context of climate change. Our proof-of-concept experiment can potentially generate evidence that genotypes with associated plasticity for water use efficiency can limit transpiration rates at high VPD values. This hypothesis could be further explored in the NIFA proposal to identify geographical areas in the US in which the limited transpiration trait would benefit soybean yields under current and future climate change scenarios.

7. Citations

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Sinclair, T.R., Messina, C.D., Beatty, A., Samples, M., 2010. Assessment across the United States of the Benefits of Altered Soybean Drought Traits. Agron J 102, 475–482. https://doi.org/10.2134/AGRONJ2009.0195
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8. Budget

Research Incentive Grant Budget*

Title: Investigating transpiration responses of soybean genotypes to vapour pressure deficit for enhanced water use efficiency and drought tolerance

PIs: Elvis Eli

Project dates: 03/2024 – 02/2026

Budget ($)

	PI	Co-PI	Co-PI
SALARIES			
GRA(s)			
Staff			
Hourly, non-student	21,000		
Hourly, enrolled student			
FRINGES	1,659		
TRAVEL - Domestic	2,000		

TRAVEL - Foreign			
EQUIPMENT			
MATERIALS & SUPPLIES	1,500		
JOURNAL PUBLICATION FEES	1,500		
OTHER DIRECT COSTS			
TOTAL	27,659		

Budget Narrative:
Funds are requested to support an undergraduate visiting scholar ($21,000 base, $1,659 benefits) to assist with data collection and running the growth chamber experiments. Domestic travel funds ($2,000) are requested to present preliminary research findings at the 2024 ASA-CSSA-SSSA conference in San Antonio, TX. Materials & Supplies ($1,500) are requested for lab supplies (e.g., pots, stakes, potting mixture, bags). Funds for publication fees are also requested ($1500).

9. Current and Pending Support

NAME (List/PD #1 first)	**SUPPORTING AGENCY AND AGENCY ACTIVE AWARD/PENDING PROPOSAL NUMBER**	**TOTAL $ AMOUNT**	**EFFECTIVE AND EXPIRATION DATES**	**% OF TIME COMMITTED**	**TITLE OF PROJECT**
	Active				
Gillman, J (USDA-ARS, Columbia, MO); Fritschi, F (University of Missouri), Lin, F (University of Missouri), Smith, R (USDA-ARS, Stoneville, MS), **Elli, E** (University of Arkansas), Vieira, C (University of Arkansas)	United Soybean Board (FY24)	Total = $375,070 My share = $46,249	Oct 2023 – Sept 2024	5%	Utilizing Genes from the Soybean Germplasm Collection to Mitigate Drought Stress
Vieira, Caio, **Elli, Elvis**	Arkansas Soybean Promotion Board	Total = $250,860 My share this year = $22,950	Apr 2024 – March 2027	5%	Overcoming Soybean Yield Plateau by Leveraging Physiology-Efficient and Yield-Formation Traits
Fernandes, Samuel, Vieira, Caio C, **Elli, Elvis F**	Research Initiative Grant (RIG)	Total = $29,100 My share = $0	June 2023 -May 2025	5%	Using gradient boosting models to combine genetic with environmental information and increase prediction ability in soybean
	Pending				
Amorim, Helen, Ashworth, Amanda J, Blackstock, Joshua, Savin, Mary C, Owens, Phillip, Hays, Phillip D, O'Brien, Peter L, **Elli, Elvis**, Thomas, Andrew L, Muenich, Rebecca, Smith, Harrison	Arkansas Water Resources Center	Total = $34,935 My share = $0		5%	One Health as affected by trees and poultry litter: Integrating remote sensing for water and nutrient best management practices in karst terrains

Gillman, J (USDA-ARS, Columbia, MO); Fritschi, F (University of Missouri), Lin, F (University of Missouri), Smith, R (USDA-ARS, Stoneville, MS), **Elli, E** (University of Arkansas), Vieira, C (University of Arkansas)	United Soybean Board (FY25)	Total = $383,673 My share = 41,780	Oct 2024 – Sept 2025	5%	Utilizing Genes from the Soybean Germplasm Collection to Mitigate Drought Stress